RAJALAKSHMI ENGINEERING COLLEGE
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CS23532 – COMPUTER NETWORKS

EX 1: Study of Various Network Commands used in Linux and Windows

AIM :

To study and execute the various network commands used in Linux and Windows and to understand their purpose through hands-on practice.

PROCEDURE :

Windows commands (Command Prompt / PowerShell)

1. arp -a : Address Resolution Protocol. Shows the IP address and MAC address of the devices (such as the router) known to your computer.

2. hostname : The simplest TCP/IP command. Displays the name of your computer.

3. ipconfig /all : Displays detailed configuration information about the TCP/IP connection, including gateway, DNS, DHCP and the type of Ethernet adapter.

4. nbtstat -a : Helps solve problems with NetBIOS name resolution (NBT stands for NetBIOS over TCP/IP).

5. netstat : Network statistics. Displays the active TCP/IP connections of the computer and is used to monitor incoming and outgoing connections, routing tables and interface statistics (e.g. netstat -r).

6. nslookup : Name server lookup. Performs DNS lookups to display the IP address of a computer, the MX records or the name servers of a domain (e.g. nslookup www.google.com).

7. pathping : Unique to Windows; a combination of ping and tracert. It traces the route to the destination and then tests each router along the way to gather statistics on packet loss at each hop.

8. ping : Packet INternet Groper. Tests connectivity between two nodes using ICMP. It can be used with a hostname (ping localhost), an IP address (ping 4.2.2.2) or a domain name (ping www.facebook.com).

9. route print : Shows and manipulates the IP routing table; mainly used to set up static routes to specific hosts or networks.

Linux commands

1. ip : Shows address information, manipulates routing and displays network devices, interfaces and tunnels.

```
ip address show
ip address add 192.168.1.254/24 dev enps03
ip link set eth0 up
```

```
ip route add default via 192.168.1.254 dev eth0
ip route get 10.10.1.4
```

2. ifconfig : Configures and troubleshoots network interfaces; it has since been replaced by the ip command.

```
ifconfig
```

3. mtr : Matt's traceroute. Combines the functions of ping and traceroute and shows statistics such as response time and packet loss for every hop.

```
mtr google.com
mtr -c 10 google.com
```

4. tcpdump : Captures and displays network packets.

```
tcpdump -D
tcpdump -i eth0 -c 10
tcpdump -i eth0 host 8.8.8.8
tcpdump -i eth0 port 53
```

5. ping : Verifies IP-level connectivity to another TCP/IP computer by sending ICMP Echo Request messages and showing the round-trip times.

```
ping google.com
ping -c 10 google.com
```

6. nmcli : Manages the network connection settings of an Ethernet connection from the command line.

```
nmcli connection show
nmcli connection modify "Wired connection 1" ipv4.method auto
```

Steps

1. Open Command Prompt or PowerShell on the Windows system.
2. Type each of the Windows commands listed above and press Enter.
3. Observe and record the output of each command.
4. Open a terminal on a Linux system and execute the Linux commands listed above.

OUTPUT:

arp -a, hostname and ipconfig /all

```
PS C:\Users\libin> arp -a

Interface: 172.20.10.4 --- 0x12
  Internet Address      Physical Address      Type
  172.20.10.1           8e-33-96-07-3d-64     dynamic
  172.20.10.15          ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.56.1 --- 0x14
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.192.152.143       01-00-5e-40-98-8f     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
PS C:\Users\libin> hostname
Windows
PS C:\Users\libin> ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : Windows
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Mixed
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
   Physical Address. . . . . . . . . : 0A-00-27-00-00-14
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::be72:4add:4931:22f8%20(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.56.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
```

nbtstat -a, netstat and nslookup

```
PS C:\Users\libin> nbtstat -a

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.

PS C:\Users\libin> netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:8080         Windows:50071          ESTABLISHED
  TCP    127.0.0.1:19575        Windows:49152          TIME_WAIT
  TCP    127.0.0.1:19575        Windows:49157          TIME_WAIT
  TCP    127.0.0.1:19575        Windows:49368          TIME_WAIT
  TCP    127.0.0.1:19575        Windows:49666          TIME_WAIT
  TCP    127.0.0.1:19575        Windows:49749          TIME_WAIT
  TCP    127.0.0.1:19575        Windows:49777          TIME_WAIT
  TCP    127.0.0.1:19575        Windows:50060          TIME_WAIT
  TCP    127.0.0.1:19575        Windows:50070          TIME_WAIT
  TCP    127.0.0.1:19575        Windows:50114          TIME_WAIT
  TCP    127.0.0.1:19575        Windows:50140          TIME_WAIT
```

```
PS C:\Users\libin> nslookup www.google.com
Server:  UnKnown
Address:  fe80::8c33:96ff:fe07:3d64

Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:482b:7700::
          2001:4860:482a:7700::
          2001:4860:4826:7700::
          2001:4860:4827:7700::
          2001:4860:482d:7700::
          2001:4860:4828:7700::
          2001:4860:4829:7700::
          2001:4860:482c:7700::
          142.251.152.119
          142.251.151.119
          142.251.150.119
          142.251.155.119
          142.251.157.119
          142.251.153.119
          142.251.154.119
          142.251.156.119
```

nslookup, pathping and ping

```
PS C:\Users\libin> nslookup www.google.com
Server:  UnKnown
Address:  fe80::8c33:96ff:fe07:3d64

Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:482b:7700::
          2001:4860:482a:7700::
          2001:4860:4826:7700::
          2001:4860:4827:7700::
          2001:4860:482d:7700::
          2001:4860:4828:7700::
          2001:4860:4829:7700::
          2001:4860:482c:7700::
          142.251.152.119
          142.251.151.119
          142.251.150.119
          142.251.155.119
          142.251.157.119
          142.251.153.119
          142.251.154.119
          142.251.156.119

PS C:\Users\libin> pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list     Loose source route along host-list.
    -h maximum_hops  Maximum number of hops to search for target.
    -i address       Use the specified source address.
    -n               Do not resolve addresses to hostnames.
    -p period        Wait period milliseconds between pings.
    -q num_queries   Number of queries per hop.
    -w timeout       Wait timeout milliseconds for each reply.
    -4               Force using IPv4.
    -6               Force using IPv6.
PS C:\Users\libin> ping localhost ping 8.8.8.8
Bad parameter ping.
PS C:\Users\libin> ping localhost

Pinging Windows [::1] with 32 bytes of data:
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms

Ping statistics for ::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

ping and route print

```
PS C:\Users\libin> ping localhost

Pinging Windows [::1] with 32 bytes of data:
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time=1ms

Ping statistics for ::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
PS C:\Users\libin> route print
===========================================================================
Interface List
 20...0a 00 27 00 00 14 ......VirtualBox Host-Only Ethernet Adapter
 21...d0 65 78 f2 fa 72 ......Microsoft Wi-Fi Direct Virtual Adapter #3
  7...d2 65 78 f2 fa 71 ......Microsoft Wi-Fi Direct Virtual Adapter #4
 18...d0 65 78 f2 fa 71 ......Intel(R) Wi-Fi 6E AX211 160MHz
 17...d0 65 78 f2 fa 75 ......Bluetooth Device (Personal Area Network)
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      172.20.10.1      172.20.10.4     45
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
      172.20.10.0  255.255.255.240         On-link       172.20.10.4    301
      172.20.10.4  255.255.255.255         On-link       172.20.10.4    301
     172.20.10.15  255.255.255.255         On-link       172.20.10.4    301
     192.168.56.0    255.255.255.0         On-link      192.168.56.1    281
     192.168.56.1  255.255.255.255         On-link      192.168.56.1    281
   192.168.56.255  255.255.255.255         On-link      192.168.56.1    281
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link      192.168.56.1    281
        224.0.0.0        240.0.0.0         On-link       172.20.10.4    301
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
```

RESULT:

Thus, the various network commands used in Linux and Windows were studied and executed successfully, and their outputs were observed.